



Unit 3 · Outcome 2 · Sustainable Development

The case study — aim & strategies

KK01

VCE Environmental Science · Units 3 & 4 · Exam study summary

Key knowledge. · the exact wording “The aim and strategies proposed for addressing the environmental science challenges associ

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.



The case study — aim & strategies

U3-O2 · KK01 Sustainable Development The case study — aim & strategies Every other dot point in this Area of Study leans on one chosen case study. This module sets it up: the real-world challenge being managed, the aim of the response, and the strategies proposed to get there — and, crucially, how to tell those three apart in an exam. Year 11 & 12 friendly 2 interactive 3D scenes 5 exam questions · weak vs strong Self-marking VCAA key knowledge · the exact wording “The aim and strategies proposed for addressing the environmental science challenges associ

What “a case study” actually means here

01 The idea What “a case study” actually means here Sustainability is a big, abstract idea. A case study is the trick that makes it examinable: you take one real, bounded environmental problem and use it as the testing ground for every concept in the Area of Study. In VCE Environmental Science, your teacher selects a single case study near the start of this Area of Study. You then return to it for all eight key-knowledge points: its sustainability dimensions, the principles, the challenges, a cost–benefit analysis, the stakeholder tensions, the impacts o

The Boorla Wetlands recovery, Yarrin River

02 The selected case study The Boorla Wetlands recovery, Yarrin River A teaching case study set in regional Victoria. It’s fictional, but every part of it mirrors real Murray–Darling-style water management you could meet in the exam. The river The Yarrin River is a regulated lowland river. Yarrin Dam upstream stores water for irrigation and town supply, so flow downstream is controlled by people, not just rain. The wetland The Boorla Wetlands (≈4,000 ha) are a Ramsar-listed floodplain: river red gum forest, migratory waterbirds, Murray cod and golden per

The environmental science challenge

03 The challenge The environmental science challenge The “challenge” is the problem — what is going wrong, and what makes it hard to fix. It is not the goal and it is not the actions. Pin it down precisely. The Boorla challenge — in one breath Decades of upstream water extraction, combined with a hotter, drier climate, have cut the volume and altered the timing of flows reaching the Boorla Wetlands — driving river red gum dieback, collapsed waterbird breeding, falling native fish numbers and rising salinity — while the very same water underpins the jobs,

The aim of the management response

04 The aim The aim of the management response The aim is the destination — a single, clear statement of what success would look like. A strong sustainability aim almost always has two halves : restore the environment while protecting the people who depend on it. The Boorla aim “To restore the ecological health of the Boorla Wetlands — re-establishing river red gum condition, waterbird breeding and native fish populations — while maintaining a secure and fair water supply for the Connawarra community, irrigators and Traditional Owners, both now and into t

The strategies proposed

05 The strategies The strategies proposed Strategies are the actions — the deliberate interventions chosen to move toward the aim. A good answer names the strategy and then says how it tackles the challenge . A name on its own is half an answer. 1 · Environmental flows Targeted releases from Yarrin Dam timed to mimic natural spring floods, triggering river red gum growth and waterbird breeding. 2 · Water-use efficiency Lining earthen channels, drip irrigation and metering so farms grow the same crops with less water — the savings returned to the river. 3

Flow simulator — feel the trade-off

06 Interactive · 3D Flow simulator — feel the trade-off This is the “environmental flows” strategy made tangible. Slide the dial to send more environmental water to the wetland and watch it recover — but watch what happens to the water left for the town and farms. There is no free lunch; that’s the point. 3D scene Environmental water allocation Drag to orbit · move the slider below Environmental water delivered: 120 GL/yr Wetland health 52% Water left for town & farms 76% Balanced. The wetland is slowly recovering and the district still has most of its w

Case-study anatomy — challenge, aim or strategy?

07 Interactive · skill drill Case-study anatomy — challenge, aim or strategy? The single most common exam slip is mislabelling these three. Read each statement and tag it. You’ll get instant feedback and a running score. Sort Tag each statement Check my answers Tagged 0 / 8 · score appears after you check.

Command words — what the verb is asking

◆ COMMAND WORDS

08 Exam technique Command words — what the verb is asking For this dot point you’ll mostly meet six command words. The verb tells you how much to write and in what shape. reveal.

P·E·L trainer — build a model answer

🔗 P·E·L WRITING

09 Writing P·E·L trainer — build a model answer P — Point: name your strategy / claim. E — Explain: how it works / the mechanism. L — Link: tie it back to the challenge or aim in the case study. Most full-mark sentences are one clean P·E·L. Interactive Question: “Outline one strategy and explain how it addresses the challenge at Boorla.” P · Point E · Explain L · Link Assemble my answer Reveal a model answer Your assembled answer Write each box, then assemble.



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Exam traps for this dot point

⚠ EXAM TRAPS

10 Pitfalls Exam traps for this dot point × Aim ≠ challenge Describing what's going wrong when asked for the aim. The challenge is the problem; the aim is the goal. Fix: aims usually start "To restore/maintain... while...". × Outcome ≠ strategy "The wetland got healthier" is a result, not a strategy. A strategy is an action you choose to take. Fix: a strategy can be done on a Monday — it's a verb. × Bare lists Naming a strategy with no mechanism or link earns half marks at best. Fix: name + how it works + link to the challenge. × Half an aim Giving only the

Practice questions — weak vs strong

× WEAK ANSWER

weak answer, an annotated

✓ STRONG ANSWER & MARKING

strong answer, a marking guide, and a self-mark. Your score collects in the dock at the bottom-right. Q1 Challenge & aim 2 marks Identify the environmental science challenge in the Boorla case study, and state the aim of the management response. Reveal answers & marking × Weak "The challenge is the environment is being damaged. The aim is to fix the wetland." Vague on both. "The environm



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

One-screen cheat sheet

✦ RECAP / CHEAT SHEET

12 Revision One-screen cheat sheet KK01 · Case study overview — at a glance Case study One bounded, real-world environmental problem used as the testing ground for the whole Area of Study. Challenge The problem : what's going wrong + what makes it hard. At Boorla: wetland decline + human dependence on the same water. Aim The goal / desired outcome. One statement, usually two-sided: "restore X while maintaining Y, now & future." Strategy A specific action to reach the aim. There are usually several. Answer = name + mechanism + link . Boorla aim Restore ri

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

The challenge is the environment is being damaged. The aim is to fix the wetland.” Vague on both. “The environment is damaged” doesn’t name the conflict between ecology and water use, and “fix the wetland” drops the human half of the aim. ≈0–1 mark.

✓ STRONG / MODEL ANSWER

The challenge is that reduced river flows (from extraction and a drying climate) are degrading the Boorla Wetlands, yet the same water sustains Connawarra’s farms, town supply and the Boorla people’s cultural water. The aim is to restore the wetland’s ecological health while keeping water supply secure and fair, now and into the future.” Names the two-sided conflict (1) and a two-sided aim (1). 2/2.

✗ WEAK ANSWER

Use less water, plant trees, and recycle. Planting trees is good for the environment.” Three loosely-named ideas, but the explanation is generic (“good for the environment”) with no mechanism and no link to the challenge. ≈1 mark.

✓ STRONG / MODEL ANSWER

Three strategies: (i) environmental flows, (ii) water-use efficiency upgrades and (iii) water buy-backs. Environmental flows deliver timed dam releases that mimic natural spring floods; this re-triggers river red gum growth and waterbird breeding that had stopped because the wetland no longer floods — directly reversing the ecological decline at the heart of the challenge.” Three valid named strategies (1) + a clear name-mechanism-link explanation of one (2). 3/3.

✗ WEAK ANSWER

The aim and the strategy are both about fixing the wetland. The aim is to fix it and the strategy is also to fix it.” No real distinction drawn, and the examples just repeat “fix it”. “Distinguish” needs a clear contrast on a shared feature. ≈1 mark.

✓ STRONG / MODEL ANSWER

An aim is the overall goal — the single statement of what success looks like — whereas a strategy is one specific action chosen to move toward that aim. At Boorla, the aim is to restore wetland health while keeping water secure and fair; a strategy such as environmental flows is just one of several actions used to achieve it. There is one aim but several strategies, and a strategy can succeed or fail without changing the aim itself.” Defines each (2), contrasts goal-vs-action and one-vs-many (1), and grounds both in Boorla examples (1). 4/4.

✗ WEAK ANSWER

They want to help pelicans. They will flood the lakes and fix the channels.” Catches one half of the aim (pelicans) but drops the “while protecting the district” half, and the strategies are named without describing how they work. ≈2 marks.

✓ STRONG / MODEL ANSWER

Aim: to restore pelican breeding at the Marra Lakes while protecting the fruit-growing district that shares the channel. Strategy 1 — a managed flood every two years using stored water, which re-creates the periodic high water levels pelicans need to breed. Strategy 2 — piping 1,200 ha of open channel, which cuts evaporation losses so the same water stretches further between the lakes and the growers.” Two-sided aim summarised (2) + two strategies each described with a mechanism (1.5 + 1.5). 5/5.

✗ WEAK ANSWER

A case study is useful because it’s a real example. The Boorla aim is to fix the wetland and the strategy is environmental flows, which adds water.” Gives a thin reason for case studies, a one-sided aim and a single strategy with no link back to the challenge. Misses the connective tissue the question asks for. ≈2–3 marks.

✓ STRONG / MODEL ANSWER

A single case study makes abstract sustainability ideas concrete and bounded, so concepts like equity and trade-offs can be analysed against real constraints and consistently across every dot point. At Boorla, the challenge is that reduced flows are degrading a Ramsar wetland while the same water sustains a town, farms and cultural sites. The aim — restore wetland health while keeping supply secure and fair — directly mirrors that two-sided challenge. The strategies then split the work: environmental flows and salinity management repair the ecology, while efficiency upgrades, recycled water and buy-backs free up the water to do so without stripping it from the community — so the strategies a